

5	(a) when waves overlap / meet the resultant displacement is the sum of the individual displacements of the waves	B1 B1	[2]
(b)	(i) 1. phase difference = $180^\circ / (n + \frac{1}{2}) 360^\circ$ (allow in rad)	B1	[1]
	2. phase difference = $0 / 360^\circ / (n360^\circ)$ (allow in rad)	B1	[1]
	(ii) $v = f\lambda$ $\lambda = 320 / 400 = 0.80 \text{ m}$	C1 A1	[2]
	(iii) path difference = $7 - 5 = 2 \text{ (m)}$ $= 2.5\lambda$	M1	
	hence minimum or maximum if phase change at P is suggested	A1	[2]